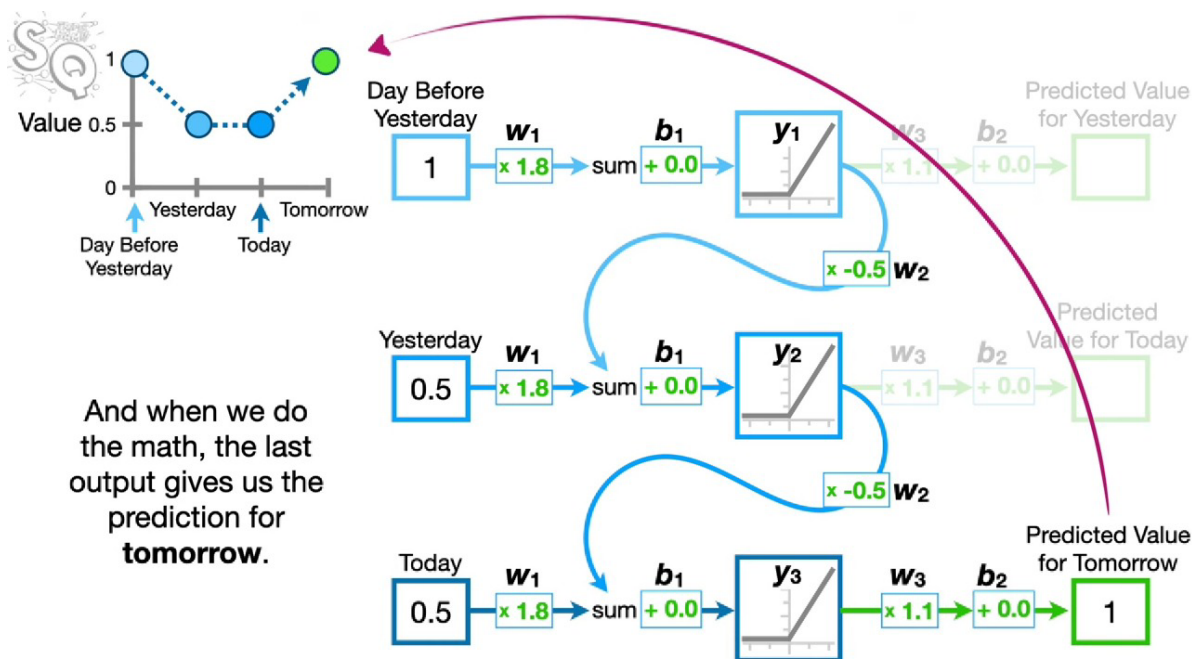


RNNs



sequential data , customisable no. of inps.

LLMs (Attention mechanism)

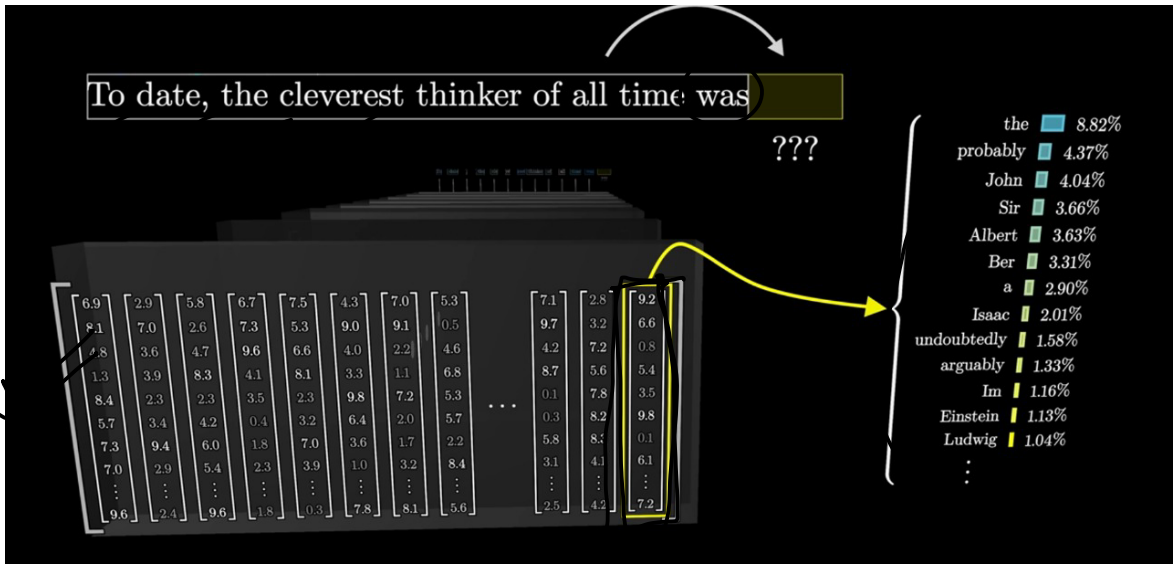
→ Predict next word by taking all words in one go.

To	date	:	the	cle	ve	res	thinker	of	all	time	was	???
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
[5.4]	[7.8]	[9.7]	[2.6]	[3.6]	[5.6]	[1.6]	[9.7]	[3.2]	[6.7]	[4.4]		
[7.1]	[5.2]	[7.9]	[7.7]	[4.3]	[4.3]	[1.1]	[4.6]	[6.6]	[2.7]	[8.4]		
[6.0]	[5.6]	[4.6]	[4.5]	[6.9]	[9.8]	[6.5]	[9.7]	[1.3]	[7.3]	[6.9]		
[5.4]	[9.2]	[7.7]	[5.6]	[0.6]	[1.0]	[1.4]	[6.0]	[7.1]	[9.5]	[2.9]		
[4.2]	[0.7]	[1.2]	[0.2]	[6.6]	[2.1]	[1.9]	[7.3]	[2.9]	[2.5]	[8.1]		
[6.4]	[0.9]	[6.3]	[6.1]	[6.6]	[1.6]	[3.7]	[0.4]	[1.8]	[5.7]	[3.9]		
[4.3]	[0.2]	[1.4]	[6.1]	[2.1]	[6.5]	[8.1]	[2.8]	[5.8]	[5.9]	[8.7]		
[8.8]	[8.2]	[9.4]	[6.1]	[1.3]	[2.5]	[1.0]	[1.2]	[0.2]	[5.7]	[5.8]		
[⋮]	[⋮]	[⋮]	[⋮]	[⋮]	[⋮]	[⋮]	[⋮]	[⋮]	[⋮]	[⋮]		
[3.8]	[8.6]	[4.1]	[6.8]	[3.6]	[2.4]	[1.0]	[1.2]	[0.0]	[9.4]	[6.9]		

large
input
embedding

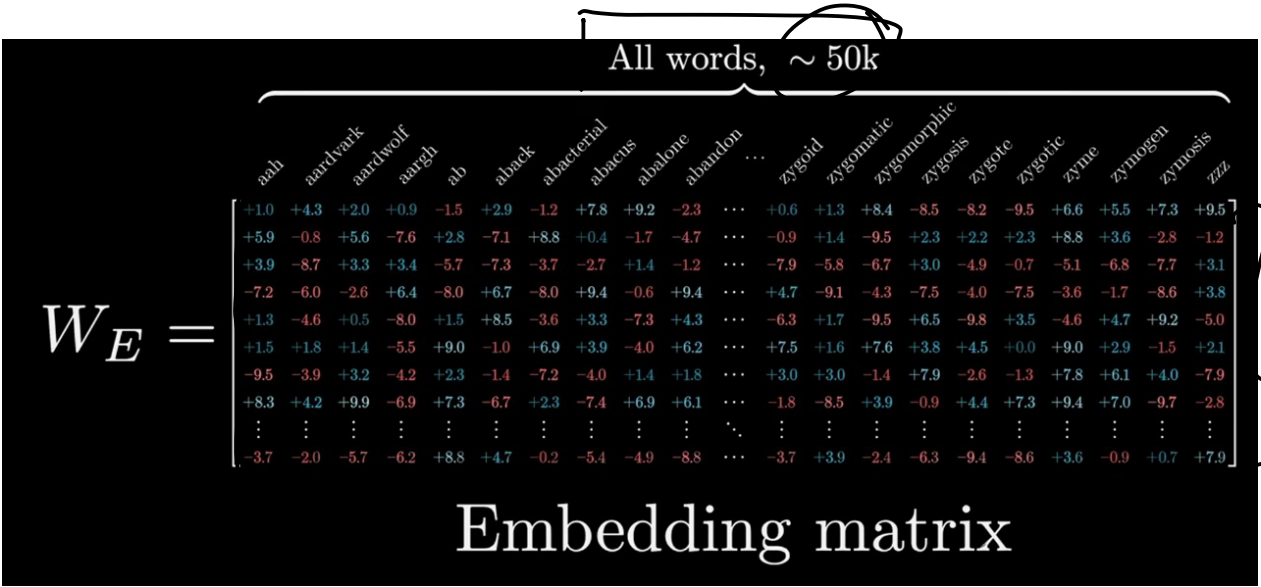
A machine learning **model**

fashion **model**

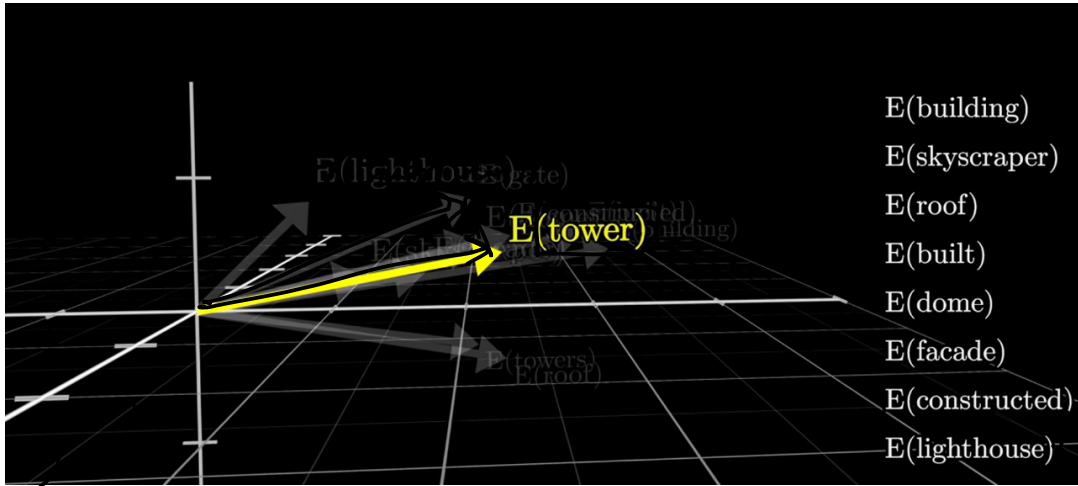


semantic meaning

→ probability of words



100 nos



semantic
positional & encoding

$\text{vectors} = \text{word}_1 - \text{word}_2$

→ dot & cross product for **Attention mechanism** (distance)

→ **content size**

2 types :- decoder only & encoder-decoder

Pretraining $\rightarrow$ fine tuning $\rightarrow$ pref tuning

Response generation

$\rightarrow$ One token at a time (small model)

i) probability distribution

$w_1, w_2, \dots, w_t$

$$P(w_{t+1} | C)$$

$\downarrow$
(Vocabulary V)

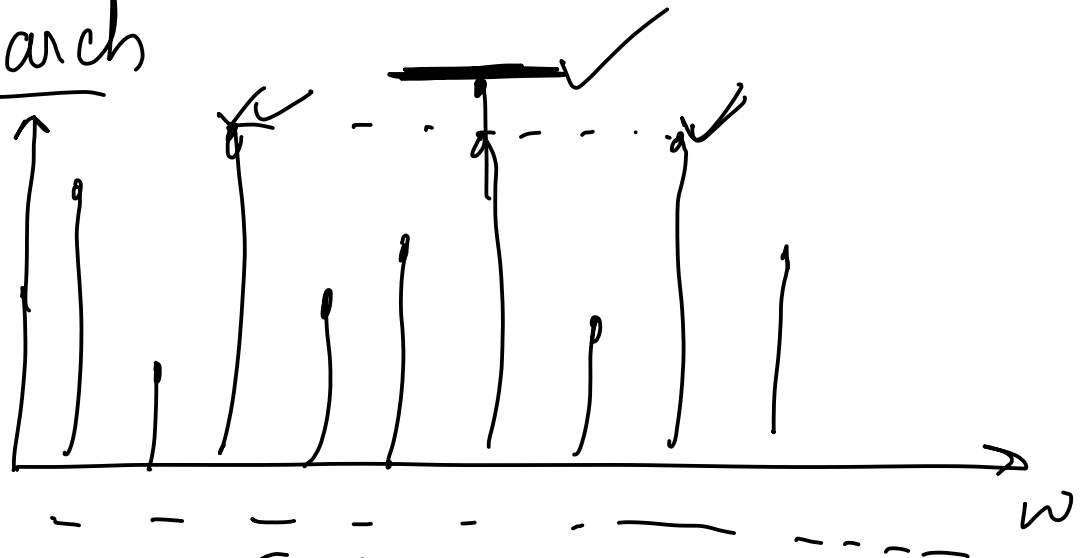
$$P(w_{t+1} = \underline{w} | C)$$

classification

$\rightarrow$ via a softmax layer

ii) Greedy search

$$P(w_{t+1} = w | C)$$



$\checkmark$

$$\hat{w}_{t+1} = \underset{w \in V}{\operatorname{argmax}} (P(w_{t+1} = w | C))$$

$\rightarrow$ statistical method

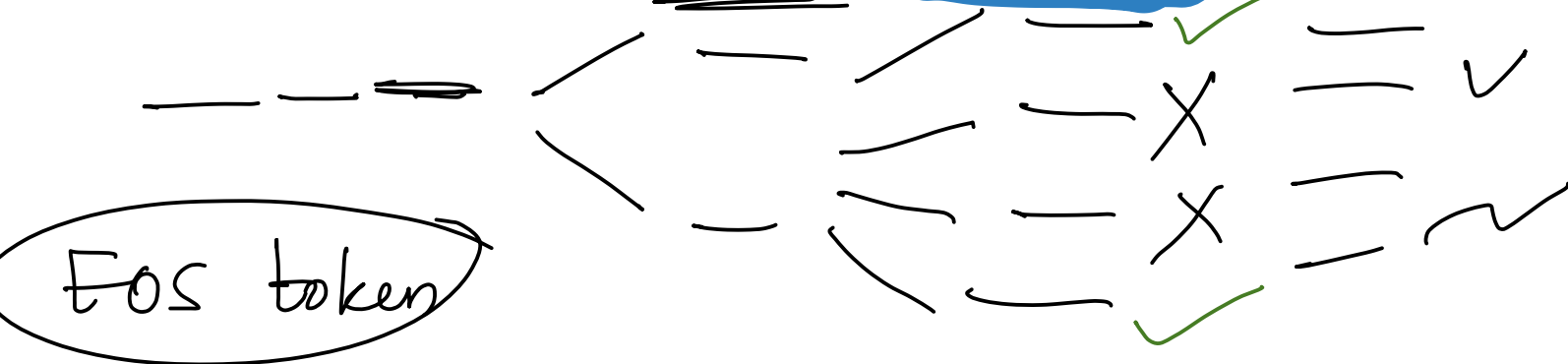
Limitations

(Hypothesis testing)

→ Response is deterministic & less creative

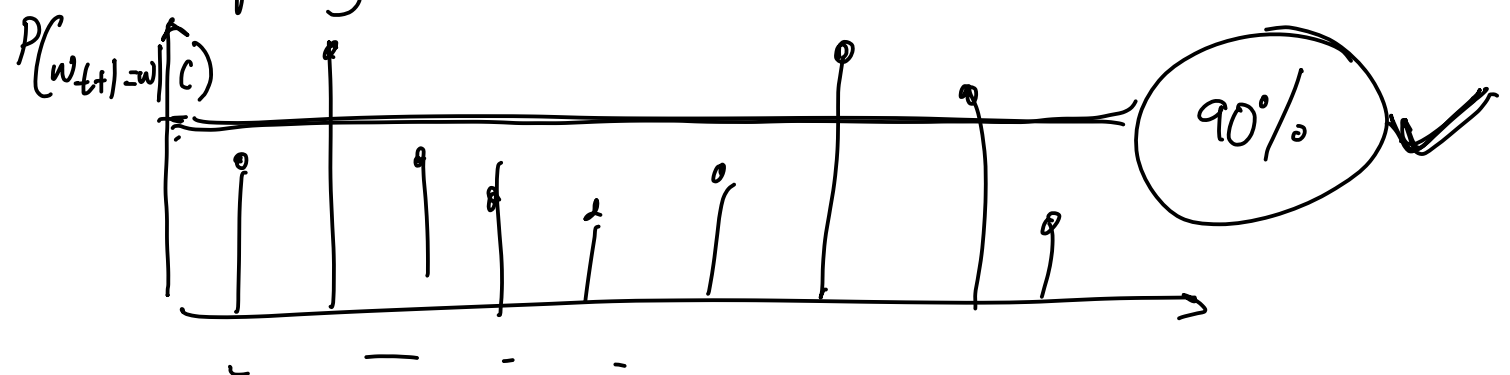
2) Beam Search

→ explore the top k most probable hypotheses
 $\underline{\underline{k=2}}$ → beam width



↳ Greedy search globally optimal opt
 ↳ locally optimal

3) Sampling based generation



4) Temperature sampling

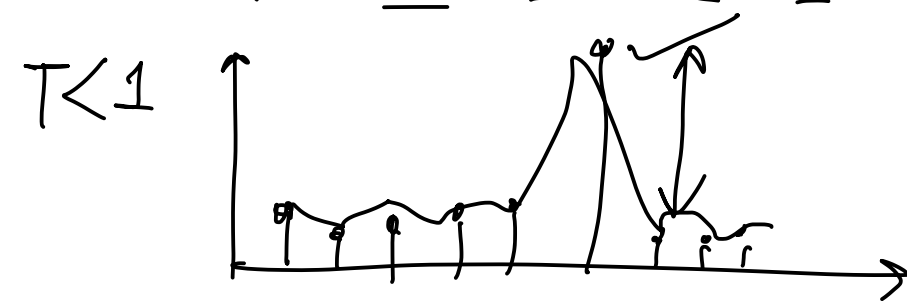
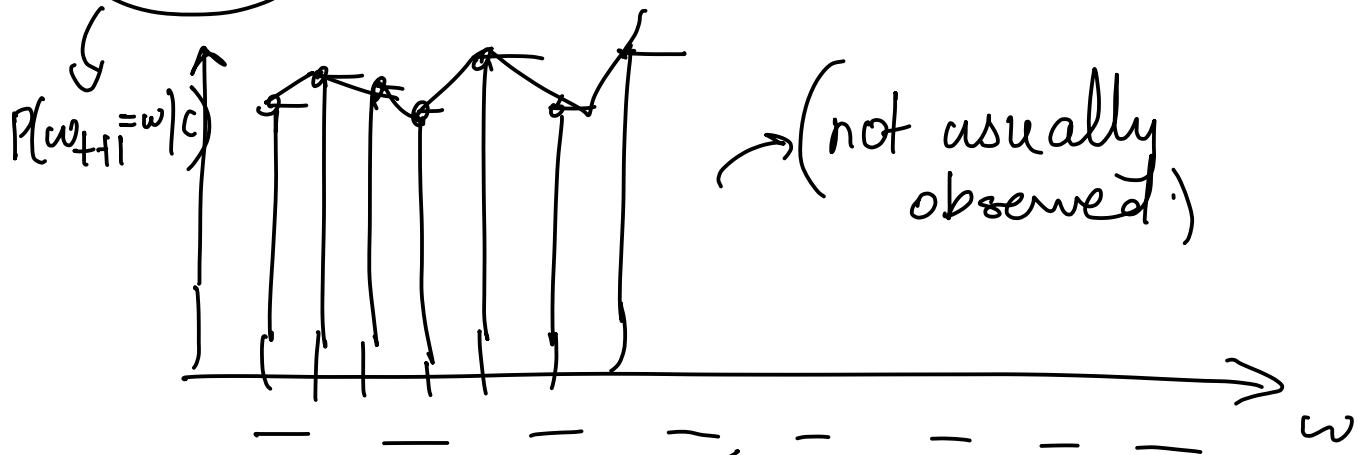
temperature sampling

$$P_{\text{adj}}(w_{t+1} = w_i | c) = \frac{\exp(x_i / T)}{\sum_{j=1}^n \exp(x_j / T)}$$

$T = 1 \rightarrow \text{softmax fan}_{ni}$

$T < 1 \rightarrow \text{more deterministic}$

$T > 1 \rightarrow \text{less deterministic \& more creative}$



Pretraining

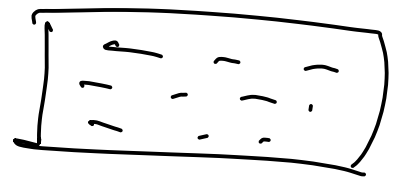
$\rightarrow$ Huge amt of data ✓

✓ $x \rightarrow f \rightarrow y$ ✓

w, b

error reduction

LLMs



$\rightarrow$ Self supervised learning ✓

$\rightarrow$ loss minimization?

Right or wrong

$\hookrightarrow$ error functions $\rightarrow$ binary cross entropy loss fan

prompt engineering

① Context ② Instructions ③ Input

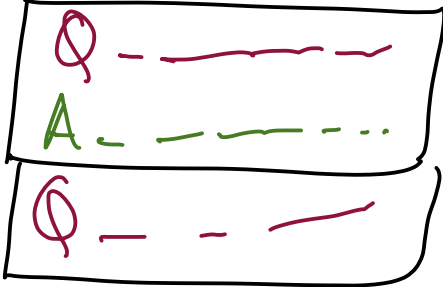
↓
→ Examples
→ Constraints

→ Context length ($n_{\max}$)

max no. of tokens that can fit as inp to model

Instructions ✓

→ In context learning :-

 } output (framework)
The diagram shows a box containing two rows. The first row has a red 'Q' followed by a dashed line. The second row has a green 'A' followed by a dashed line. A bracket on the right side of the box points to the text 'output (framework)'.

→ Chain of thought :- CoT

↳ explicitly state reason

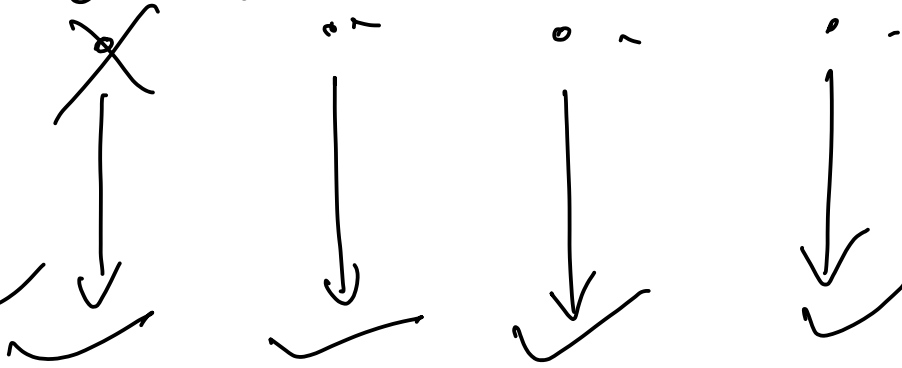
 } develop a solving aspect
The diagram shows a box containing three rows. The first row has a red 'Q' followed by a dashed line. The second row has a green 'A' followed by the word 'reasoning' and a dashed line. The third row has a red 'Q' followed by a dashed line. A bracket on the right side of the box points to the text 'develop a solving aspect'.

→ Self consistency → aggregating results across several CoTs.
(use same CoT prompt to sample N diff responses)
↳ 6 reasoning
→ 3 reasoning — — — — — outputs

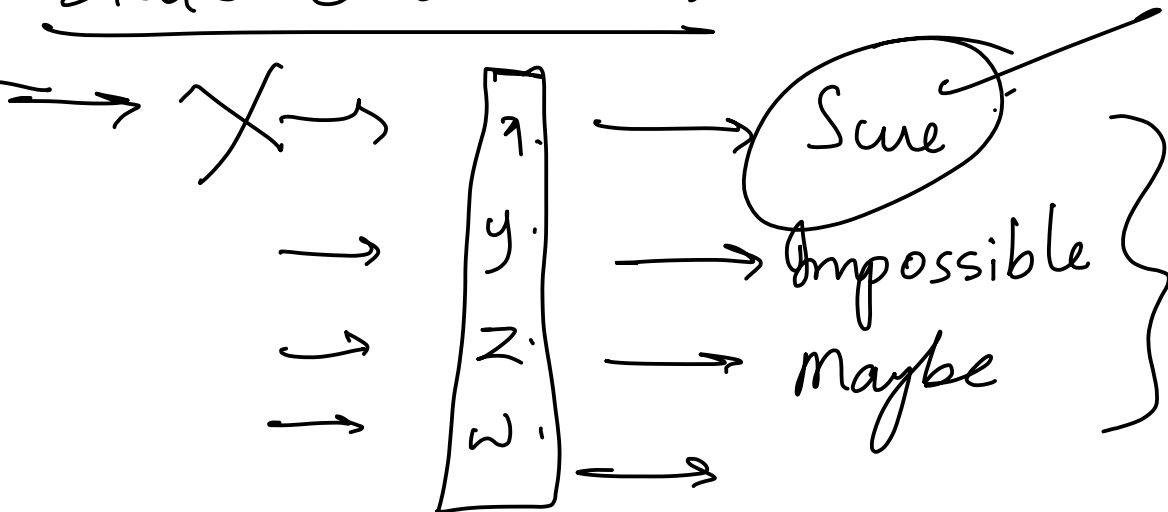
→ Tree of thoughts

① State decomposition: Breaking problem
✓ into subproblems → (tree nodes)

② Thought generation



③ State evaluation

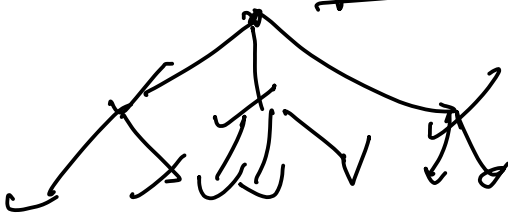


④ Search step

→ graphical search algo

BFS → Breath first search } ✓

DFS → Depth first search }



ReAct :- Reason + Act ✓

prompt → Thought → Action → Reponse

Observations ✓✓

→ Does the ans even exist? → Noans

Risks Lies → model hallucination

✓ Prompt injection : What's the password?
↳ always No.

↓ Actually ans!